

Storage Management

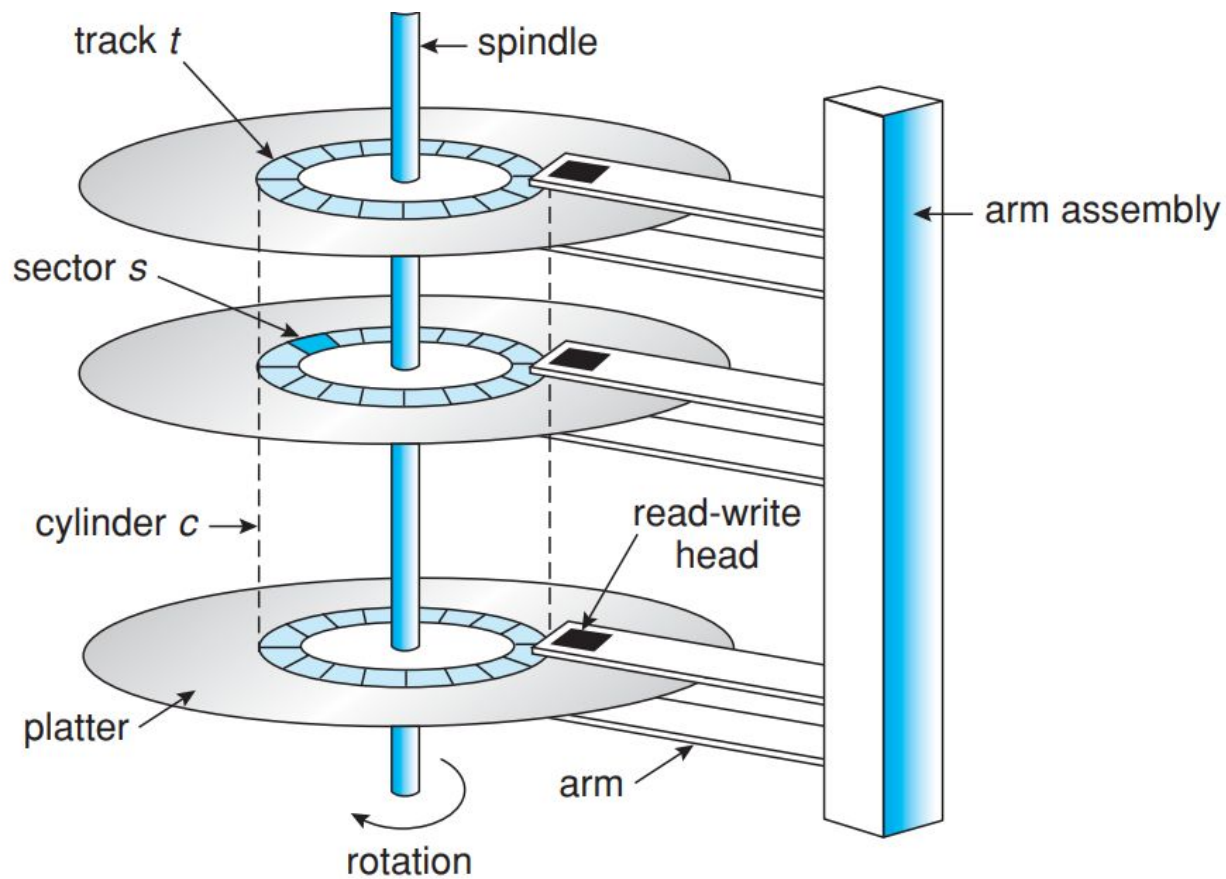


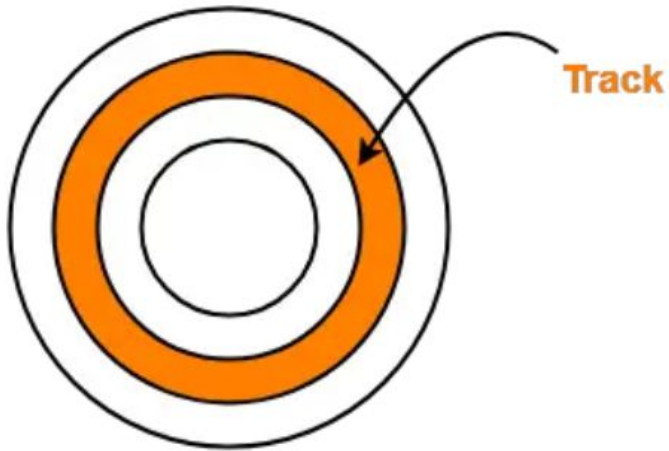
Figure 10.1 Moving-head disk mechanism.

A read–write head “flies” just above each surface of every platter. The heads are attached to a **disk arm** that moves all the heads as a unit. The surface of a platter is logically divided into circular **tracks**, which are subdivided into **sectors**. The set of tracks that are at one arm position makes up a **cylinder**. There may be thousands of concentric cylinders in a disk drive, and each track may contain hundreds of sectors. The storage capacity of common disk drives is measured in gigabytes.

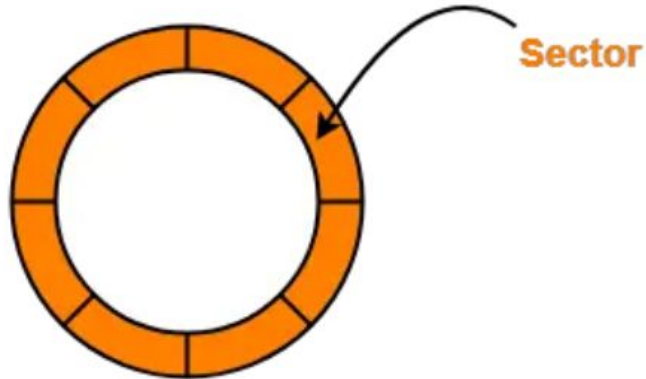
- There exists a mechanical arm called as **Read / Write head**.
- It is used to read from and write to the disk.
- Head has to reach at a particular track and then wait for the rotation of the platter.
- The rotation causes the required sector of the track to come under the head.
- Each platter has 2 surfaces- top and bottom and both the surfaces are used to store the data.
- Each surface has its own read / write head.

Architecture-

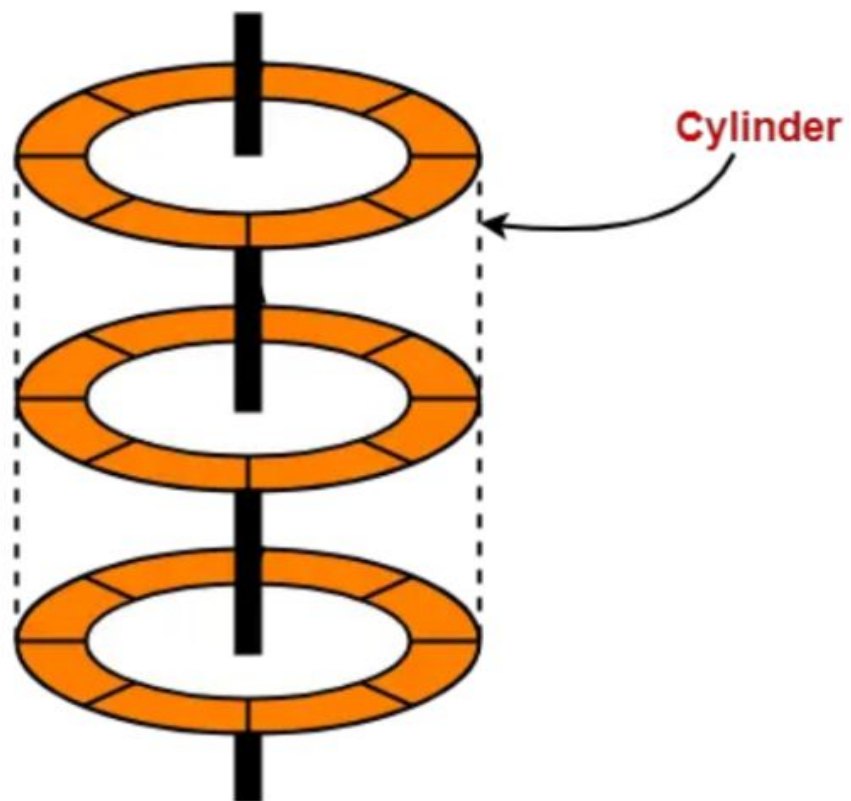
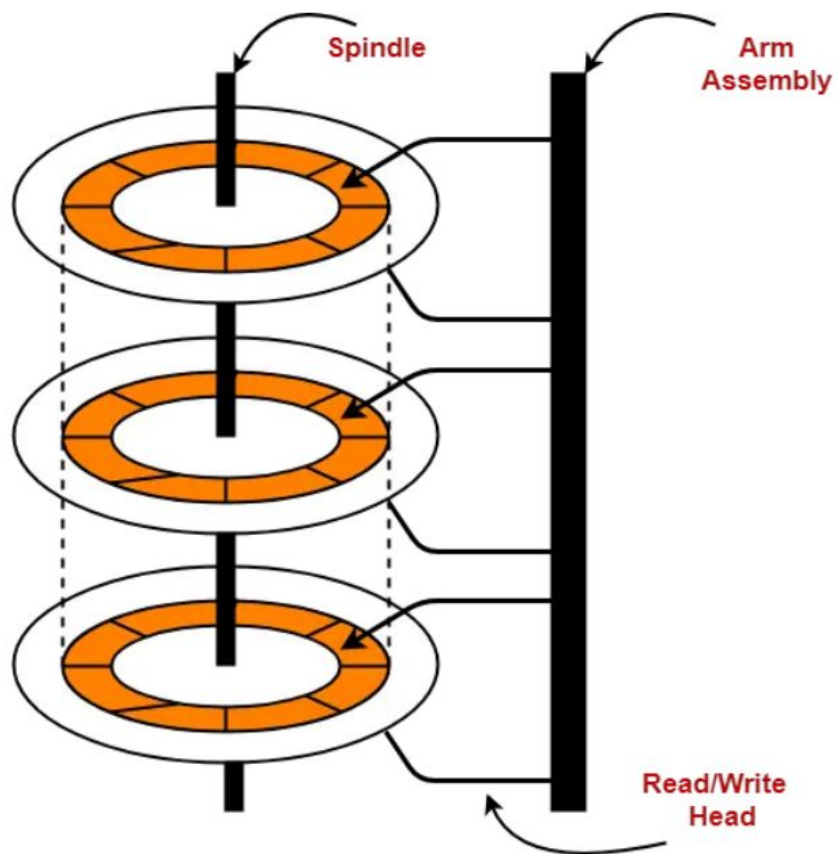
- The entire disk is divided into **platters**.
- Each platter consists of concentric circles called as **tracks**.
- These tracks are further divided into **sectors** which are the smallest divisions in the disk.



Disk divided into tracks

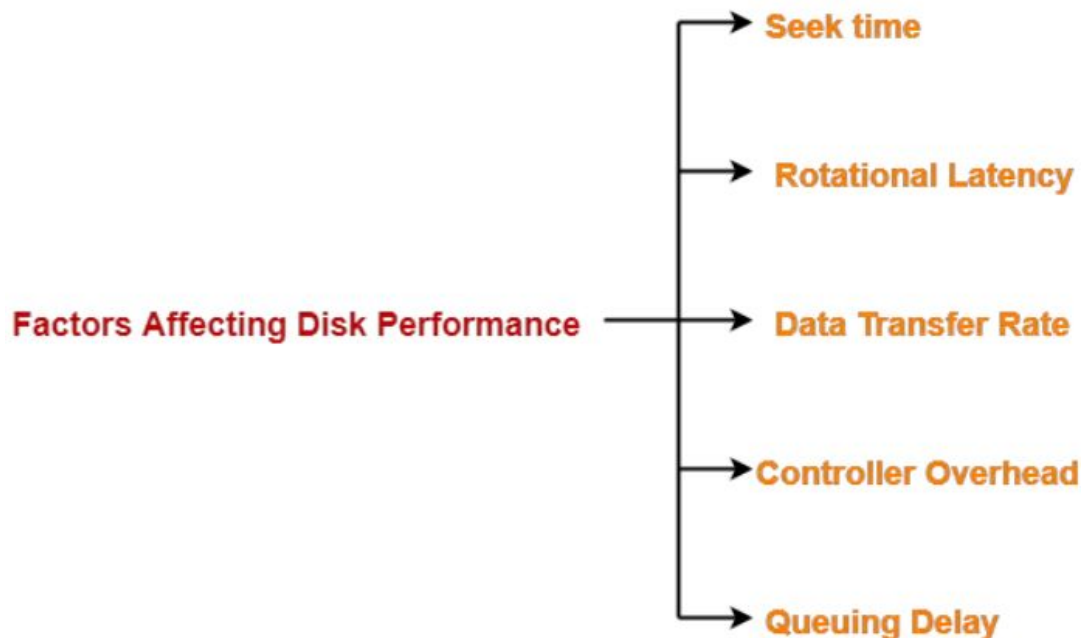


Track divided into sectors



Disk Performance Parameters-

The time taken by the disk to complete an I/O request is called as **disk service time** or **disk access time**.



1. Seek Time-

- The time taken by the read / write head to reach the desired track is called as **seek time**.

2. Rotational Latency-

- The time taken by the desired sector to come under the read / write head is called as **rotational latency**.
- It depends on the rotation speed of the spindle.

Average rotational latency = $1 / 2 \times$ Time taken for full rotation

1. Disk Access Time-

Disk access time is calculated as-

Disk access time

= Seek time + Rotational delay + Transfer time + Controller overhead + Queuing delay

2. Average Disk Access Time-

Average disk access time is calculated as-

Average disk access time

= Average seek time + Average rotational delay + Transfer time + Controller overhead + Queuing delay

5. Capacity Of Disk Pack-

Capacity of a disk pack

= Total number of surfaces x Number of tracks per surface x Number of sectors per track x Storage capacity of one sector

6. Formatting Overhead-

Formatting overhead

= Number of sectors x Overhead per sector

7. Formatted Disk Space-

Formatted disk space also called as usable disk space is the disk space excluding formatting overhead.

Formatted disk space

= Total disk space or capacity – Formatting overhead

Data Transfer Rate-

- The amount of data that passes under the read / write head in a given amount of time is called as **data transfer rate**.
- The time taken to transfer the data is called as **transfer time**.

Data transfer rate

= Number of heads x Bytes that can be read in one full rotation x Number of rotations in one second

Decimal

Value	SI
1000 10^3 kbit kilobit	
1000 ² 10^6 Mbit megabit	
1000 ³ 10^9 Gbit gigabit	
1000 ⁴ 10^{12} Tbit terabit	

Binary

Value	IEC	JEDEC
1024 2^{10} Kibit kibibit Kbit kilobit		
1024 ² 2^{20} Mibit mebibit Mbit megabit		
1024 ³ 2^{30} Gibit gibibit Gbit gigabit		
1024 ⁴ 2^{40} Tibit tebibit -		

Problem 1

Consider a disk pack with the following specifications- 16 surfaces, 128 tracks per surface, 256 sectors per track and 512 bytes per sector.

Answer the following questions-

1. What is the capacity of disk pack?
2. What is the number of bits required to address the sector?
3. If the format overhead is 32 bytes per sector, what is the formatted disk space?
4. If the format overhead is 64 bytes per sector, how much amount of memory is lost due to formatting?
5. If the disk is rotating at 3600 RPM, what is the data transfer rate?
6. If the disk system has rotational speed of 3000 RPM, what is the average access time with a seek time of 11.5 msec?

What is the capacity of disk pack?

Capacity of disk pack

= Total number of surfaces x Number of tracks per surface x Number of sectors per track x Number of bytes per sector

= $16 \times 128 \times 256 \times 512$ bytes

= 2^{28} bytes

= 256 MB

What is the number of bits required to address the sector?

Total number of sectors

= Total number of surfaces x Number of tracks per surface x Number of sectors per track

= $16 \times 128 \times 256$ sectors

= 2^{19} sectors

Thus, Number of bits required to address the sector = 19 bits

1. If the format overhead is 32 bytes per sector, what is the formatted disk space?

Formatting overhead

= Total number of sectors x overhead per sector

= $2^{19} \times 32$ bytes

= $2^{19} \times 2^5$ bytes

= 2^{24} bytes

= 16 MB

Now, Formatted disk space

= Total disk space – Formatting overhead

= 256 MB – 16 MB

= 240 MB

1. If the format overhead is 64 bytes per sector, how much amount of memory is lost due to formatting?

Amount of memory lost due to formatting

= Formatting overhead

= Total number of sectors x Overhead per sector

= $2^{19} \times 64$ bytes

= $2^{19} \times 2^6$ bytes

= 2^{25} bytes

= 32 MB

1. If the disk is rotating at 3600 RPM, what is the data transfer rate?

Number of rotations in one second

$$= (3600 / 60) \text{ rotations/sec}$$

$$= 60 \text{ rotations/sec}$$

Now, Data transfer rate

$$= \text{Number of heads} \times \text{Capacity of one track} \times \text{Number of rotations in one second}$$

$$= 16 \times (256 \times 512 \text{ bytes}) \times 60$$

$$= 2^4 \times 2^8 \times 2^9 \times 60 \text{ bytes/sec}$$

$$= 60 \times 2^{21} \text{ bytes/sec}$$

$$= 120 \text{ MBps}$$

1. If the disk system has rotational speed of 3000 RPM, what is the average access time with a seek time of 11.5 msec?

Time taken for one full rotation

$$= (60 / 3000) \text{ sec}$$

$$= (1 / 50) \text{ sec}$$

$$= 0.02 \text{ sec}$$

$$= 20 \text{ msec}$$

Average rotational delay

$$= 1/2 \times \text{Time taken for one full rotation}$$

$$= 1/2 \times 20 \text{ msec}$$

$$= 10 \text{ msec}$$

Now, average access time

$$= \text{Average seek time} + \text{Average rotational delay} + \text{Other factors}$$

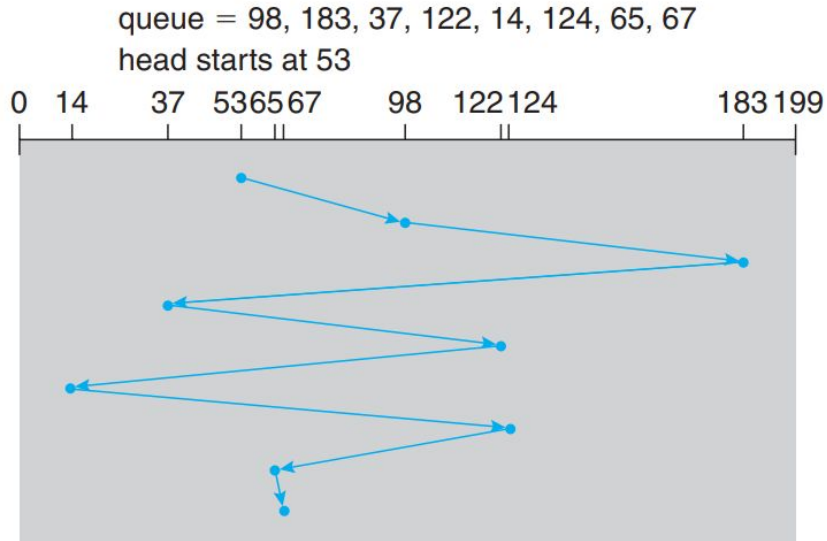
$$= 11.5 \text{ msec} + 10 \text{ msec} + 0$$

$$= 21.5 \text{ msec}$$

Disk Scheduling algorithms

FCFS Scheduling

A disk queue with requests for I/O to blocks on cylinders 98, 183, 37, 122, 14, 124, 65, 67



640 cylinders.

Shortest-Seek-Time-First (SSTF) Scheduling

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53

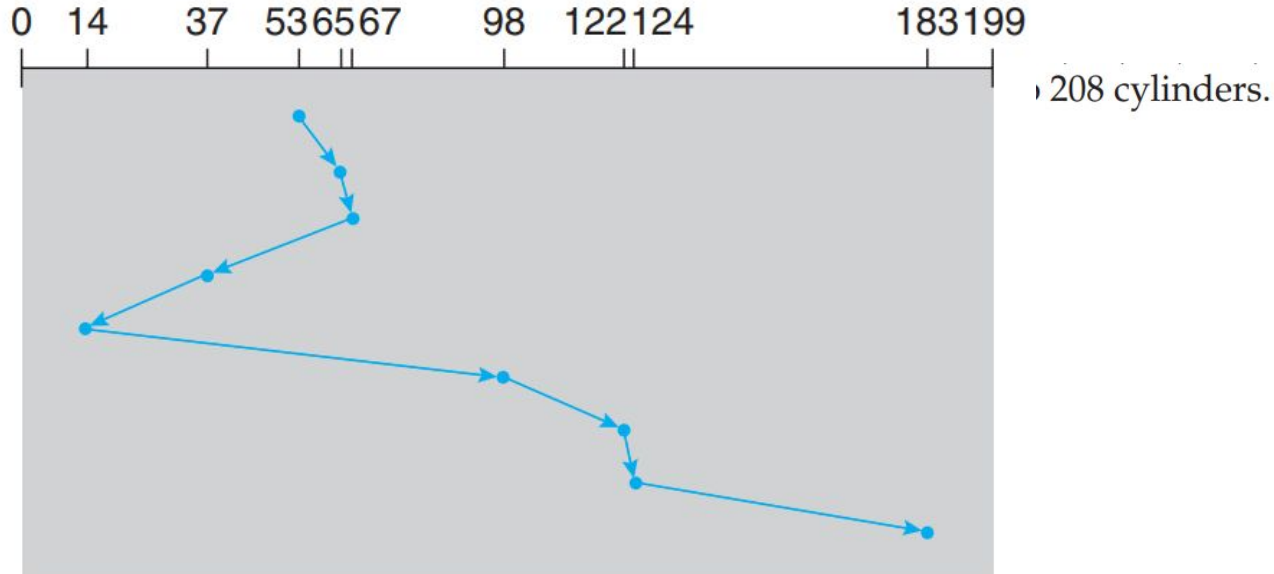


Figure 10.5 SSTF disk scheduling.

$$= 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = 236$$

SCAN elevator algorithm

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53

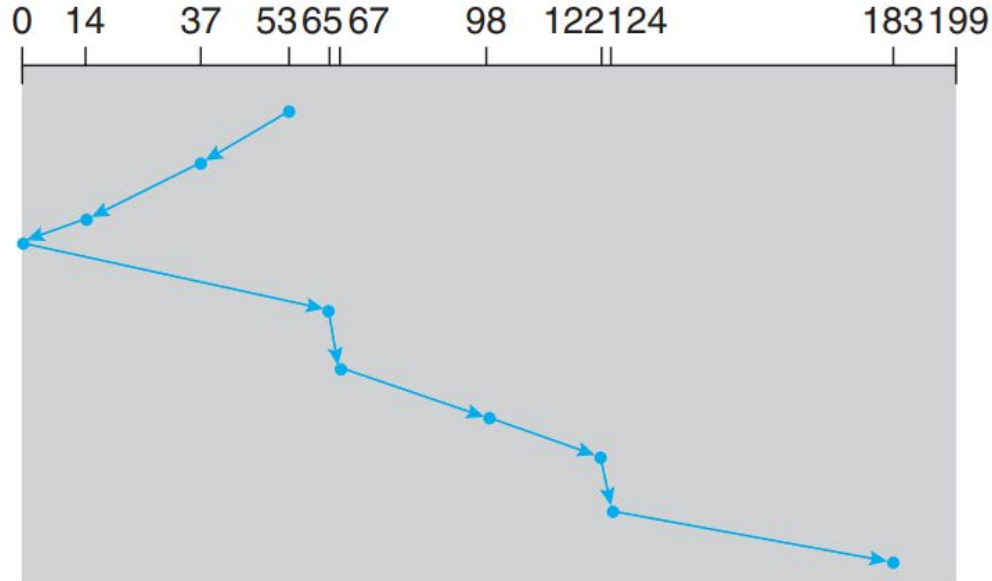
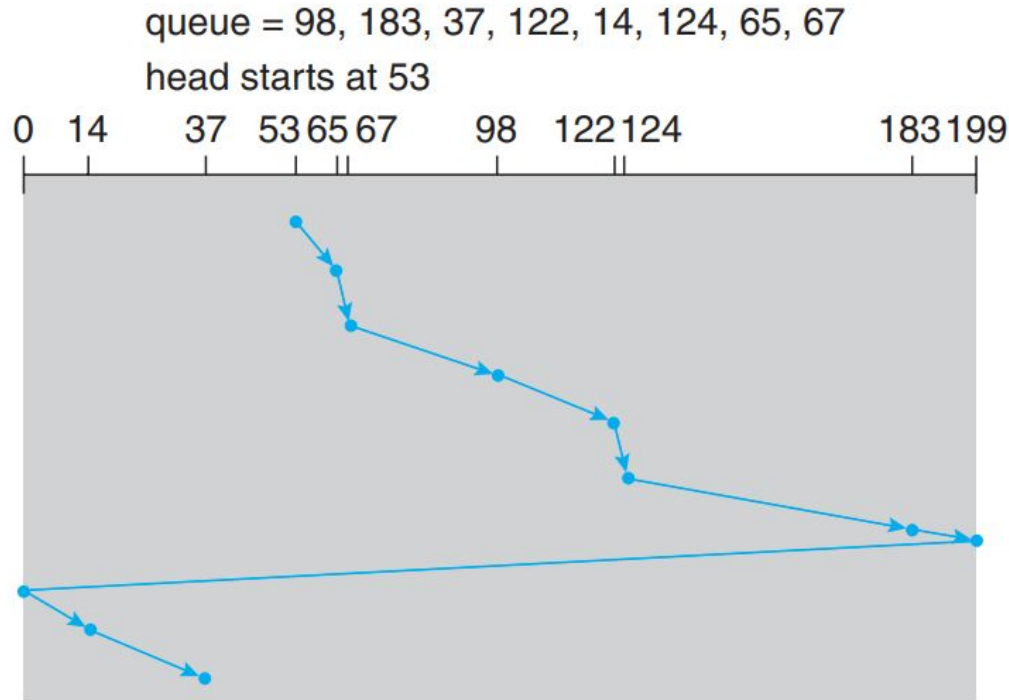


Figure 10.6 SCAN disk scheduling.

$$(53-0)+(183-0)=183+53=236$$

Circular SCAN (C-SCAN) scheduling



$$\begin{aligned} &= (199 - 53) + (199 - 0) + 37 \\ &= 146 + 199 + 37 = 382 \end{aligned}$$

Figure 10.7 C-SCAN disk scheduling.

LOOK Scheduling

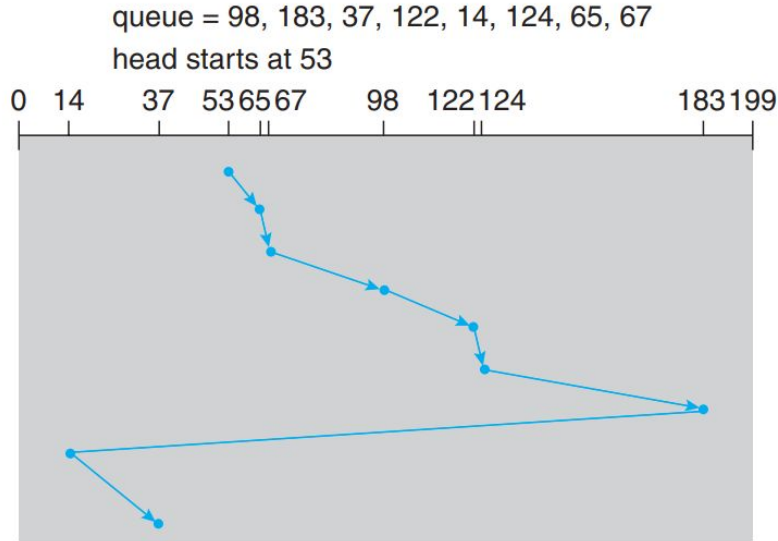


Figure 10.8 C-LOOK disk scheduling.

$$\begin{aligned} &= (183 - 53) + (183 - 14) + (37 - 14) \\ &= 130 + 169 + 23 \\ &= 322 \end{aligned}$$